# FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control

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## Motivation

### Same task. Different dynamics.

A new environment can break precise control.

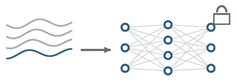
**MuJoCo** 6 s → 10 s → 14 s

**Real world · sand** **Unstable execution**

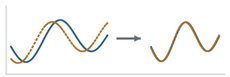
**Can a few self-collected rollouts restore precise control?**

## Prior approaches

**Domain randomization [1]**
Vary training dynamics.



**In-context adaptation [2]**
Fixed policy weights.

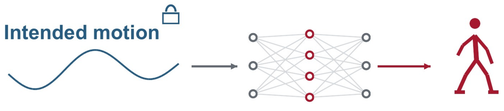


**System identification [3]**
Fit residuals; retrain.

**RL finetune [4]**
Reinforcement learning.

## FADA: update policy weights

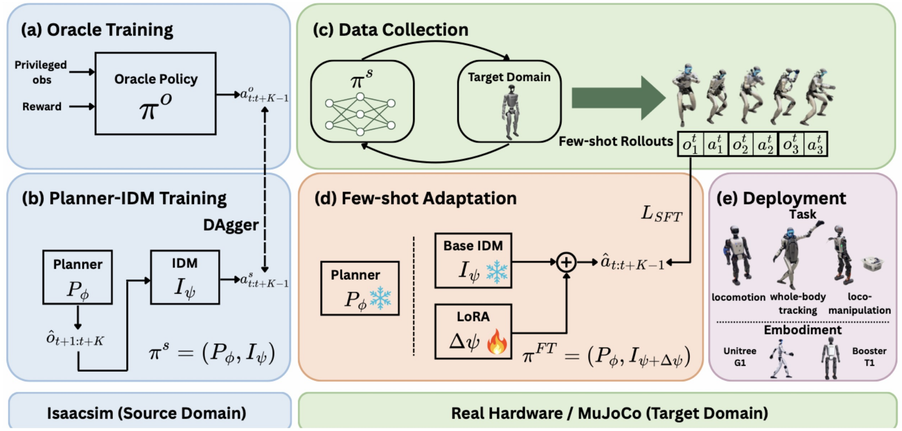
From self-collected rollouts.
Keep the planner fixed; adapt the IDM.




**IDM + LoRA**

**~2 min of target rollouts**
No target rewards, expert demonstrations or simulator fitting.

## FADA framework

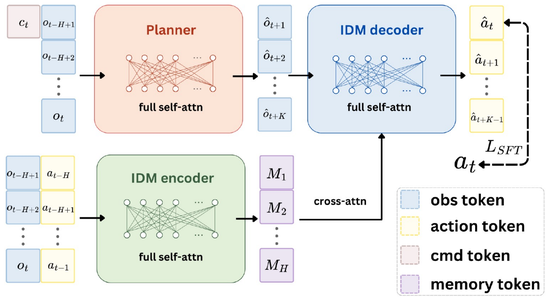



$$\widehat{Y}_t^K = P_\phi(\mathcal{O}_t^H, c_t), \qquad a_t = \Pi_1 I_\psi\left(h_t, \widehat{Y}_t^K\right)$$

$\pi^o$ : oracle policy; $\pi^s$ : source-trained student; $\pi^{FT}$ : target-adapted policy.

## Architecture

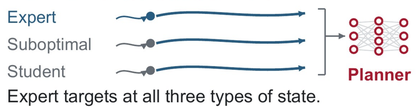
DAgger: dataset aggregation with expert relabels




### IDM: diverse behavior

All behavior + expert-relabelled continuations.

### Planner: expert supervision




Expert targets at all three types of state.

### Models

$P_\phi$ : planner
$I_\psi$ : inverse dynamics model
$\phi, \psi$ : model weights
$\bar{\psi}$ : frozen IDM weights
$\Delta\psi$ : low-rank update
$\Pi_1$ : first action in a chunk
$M$ : encoded history tokens

### Observations & actions

$o$ : body/joint observation
$a$ : executed; $a^\star$ : expert action
$\hat{o}, \hat{a}$ : predicted values
$\widehat{Y}$ : predicted future
$Y_{\text{exec}}$ : observed future
$\mathcal{O}, \mathcal{A}$ : obs/action histories
$c$ : task command

### History & losses

$h = (\mathcal{O}, \mathcal{A})$ : input history
$t$ : time step
$H, K$ : history/future lengths
$\mathbb{E}_{\text{src/tgt}}$ : source/target mean
$\mathcal{L}$ : mean squared action error
$L_{SFT}$ : supervised fine-tuning loss
attn: attention; obs: observation

### Train IDM

Use all behavior, including suboptimal rollouts.

$$\mathcal{L}_I(\psi) = \mathbb{E}_{\text{src}}\left[\left\| \Pi_1 I_\psi(h_t, Y^K_{t,\text{exec}}) - a_t \right\|_2^2\right]$$

### Train planner

Use expert action targets; hold IDM weights fixed.

$$\mathcal{L}_P(\phi) = \mathbb{E}_{\text{src}}\left[\left\| \Pi_1 I_{\bar{\psi}}\left(h_t, P_\phi(\mathcal{O}_t^H, c_t)\right) - a_t^\star \right\|_2^2\right]$$

## Target adaptation

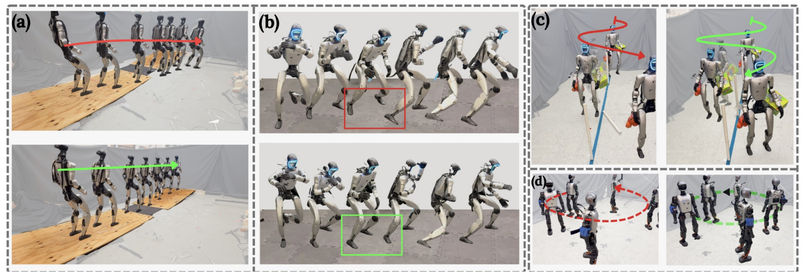
**~2 min** of self-collected rollouts.
Freeze planner + base IDM.
Train **LoRA**: a low-rank adapter.

$$\widehat{a}_t = \Pi_1 I_{\psi+\Delta\psi}(h_t, Y^K_{t,\text{exec}})$$

$$\mathcal{L}_{\text{adapt}} = \mathbb{E}_{\text{tgt}}\left[\|\widehat{a}_t - a_t\|_2^2\right]$$

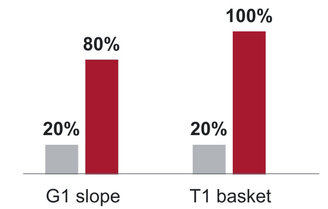
Fit observed future-action pairs.

## Results: few-shot hardware adaptation




Unitree G1: slope, terrain, groceries. Booster T1: payload.
Red: zero-shot failures. Green: after IDM adaptation.

**Task success ↑** ■ FADA zero-shot ■ FADA adapted




TF–DAgger: 0% on both tasks

**Normalized tracking error ↓**

TF–DAgger = 1.0 (dashed)
Means over 5 trials per task; paper Table 1.

Across 2 completion tasks: **20%** → **90%** mean success.
Across 3 tracking tasks: **27.4% lower error** vs. zero-shot.

## Analysis: what should be adapted?

**IsaacSim → MuJoCo: normalized task error ↓**

| | FADA zero-shot | FADA adapted | CoPred finetuned |
|---|---|---|---|
| G1 slope | 0.946 | 0.800 | 1.611 |
| G1 Kungfu | 0.961 | 0.714 | 1.421 |
| T1 payload | 1.042 | 0.885 | 1.737 |
| T1 slope | 1.034 | 0.914 | 1.193 |
| T1 Falcon (force) | 0.880 | 0.347 | 1.054 |

**TF–DAgger:** source transformer trained by oracle imitation.
**CoPred:** joint observation/action predictor; future-observation finetuning.

**24.7% lower error** vs. zero-shot; CoPred worsens all 5 tasks.
20 trials/task; ~2 min adaptation; TF–DAgger = 1.0 (Table 2).

**Why LoRA?** Full IDM finetuning raises error 10–29%; LoRA lowers it 12–15% vs. zero-shot on 3 tasks (Table 4).

**Limitations** Needs usable rollouts; task-specific IDM; body sensors only.

**References**
[1] Gu et al. Humanoid-Gym. arXiv:2404.05695, 2024.
[2] Kumar et al. RMA. RSS 2021.
[3] He et al. ASAP. RSS 2025.
[4] Smith et al. Legged Robots that Keep on Learning. 2021.